

Civics Group	Index Number	Name (use BLOCK LETTERS)
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ST. ANDREW'S JUNIOR COLLEGE
2008 JC2 Prelims

H2 BIOLOGY

9747/2

Paper 2: Core

Wednesday

27 August 2008

2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all seven** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Questions)

Choose **one** of the two questions.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,

1. Attach Section B answers to the **cover sheet** provided.
2. **Enter the number** of the Section B questions you have answered on the cover sheet.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/12
2	/9
3	/12
4	/9
5	/13
6	/13
7	/12
Total	/80

This document consists of **19** printed pages.

[Turn over

Section A

Answer **all** questions.

QUESTION 1

In humans a gene coding for a certain protein is located on chromosome 12. The gene contains 1125 base pairs, but the mature mRNA strand only contains 456 bases. The protein contains 140 amino acids.

Fig. 1.1 shows seven triplets that code for this protein.

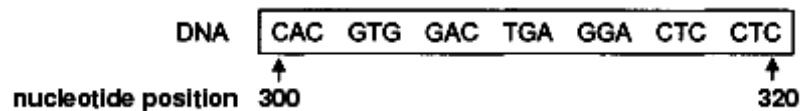


Fig. 1.1

(a) (i) State the total number of bases that code for the amino acids in this protein? [1]

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(ii) Using your answer in (a)(i), explain the number of bases in the gene and the mature mRNA. [2]

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Table 1.2 lists the amino acids and their corresponding mRNA codons.

UUU	phe	UCU	ser	UAU	tyr	UGU	cys
UUC	phe	UCC	ser	UAC	tyr	UGC	cys
UUA	leu	UCA	ser	UAA	STOP	UGA	STOP
UUG	leu	UCG	ser	UAG	STOP	UGG	trp
CUU	leu	CCU	pro	CAU	his	CGU	arg
CUC	leu	CCC	pro	CAC	his	CGC	arg
CUA	leu	CCA	pro	CAA	gln	CGA	arg
CUG	leu	CCG	pro	CAG	gln	CGG	arg
AUU	ileu	ACU	thr	AAU	asn	AGU	ser
AUC	ileu	ACC	thr	AAC	asn	AGC	ser
AUA	ileu	ACA	thr	AAA	lys	AGA	arg
AUG	met	ACG	thr	AAG	lys	AGG	arg
GUU	val	GCU	ala	GAU	asp	GGU	gly
GUC	val	GCC	ala	GAC	asp	GGC	gly
GUA	val	GCA	ala	GAA	glu	GGA	gly
GUG	val	GCG	ala	GAG	glu	GGG	gly

Table 1.2

- (b)** Using **Table 1.2**, determine the amino acid sequence for the portion of DNA given in **Fig. 1.1** [1]

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Two mutations are commonly associated with this gene on the 20th nucleotide; a single base deletion at position 315 or a single base substitution by adenine at the same position.

- (c) (i)** Describe the structure of a DNA polynucleotide. [3]

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- (ii)** Explain which of the above mutation is likely to be more harmful. [3]

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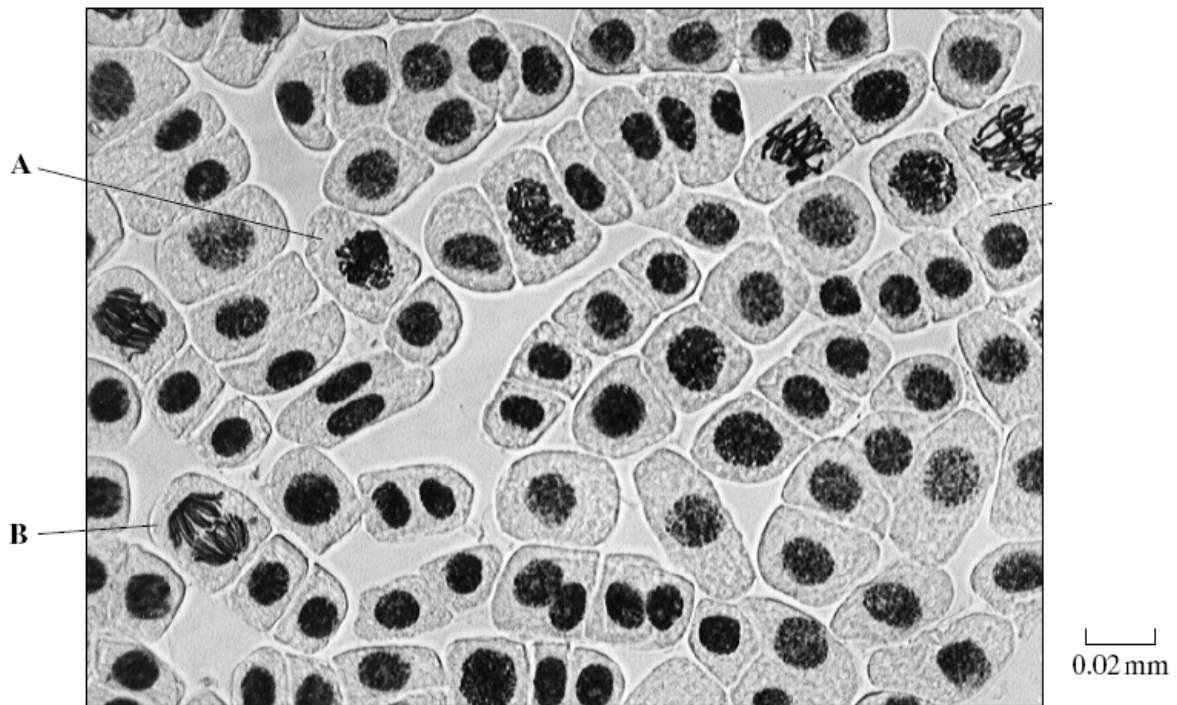
- (iii)** Explain how the change in sequence of the DNA molecule can affect the phenotype of the organism. [2]

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[Qn 1 Total: /12]

QUESTION 2

- (a) The photograph shows cells from an onion root tip. The root tip has been squashed and stained to show the stages of mitosis.



- (i) At what stage of mitosis is cell **A**? [1]

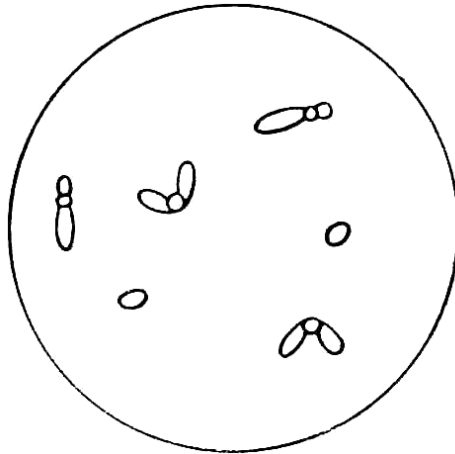
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- (ii) What is the evidence that cell **B** is in anaphase? [1]

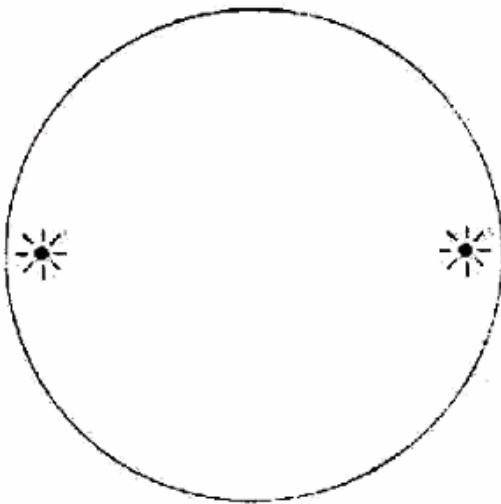
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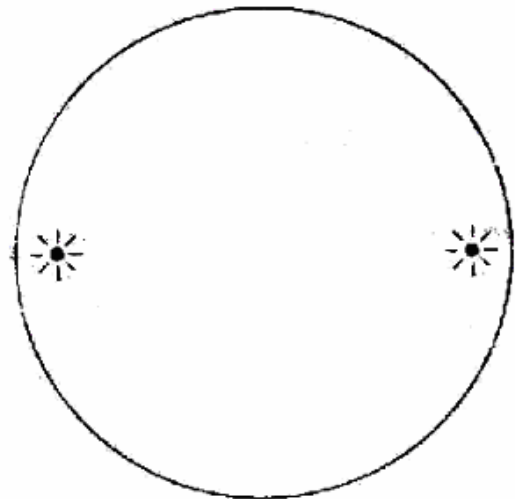
- (b) The diagram below represents the chromosomes contained in a somatic cell of an animal ($2n=6$).



Using the same chromosomes shown, complete the empty cells below to show clearly the essential differences between metaphase of mitosis and metaphase I of meiosis. [3]



Metaphase (Mitosis)



Metaphase I (Meiosis)

- (c)** Explain why haploid cells need to be produced during a lifecycle which includes sexual reproduction. [2]

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- (d)** Explain how cancer is a result of uncontrolled cell division. [2]

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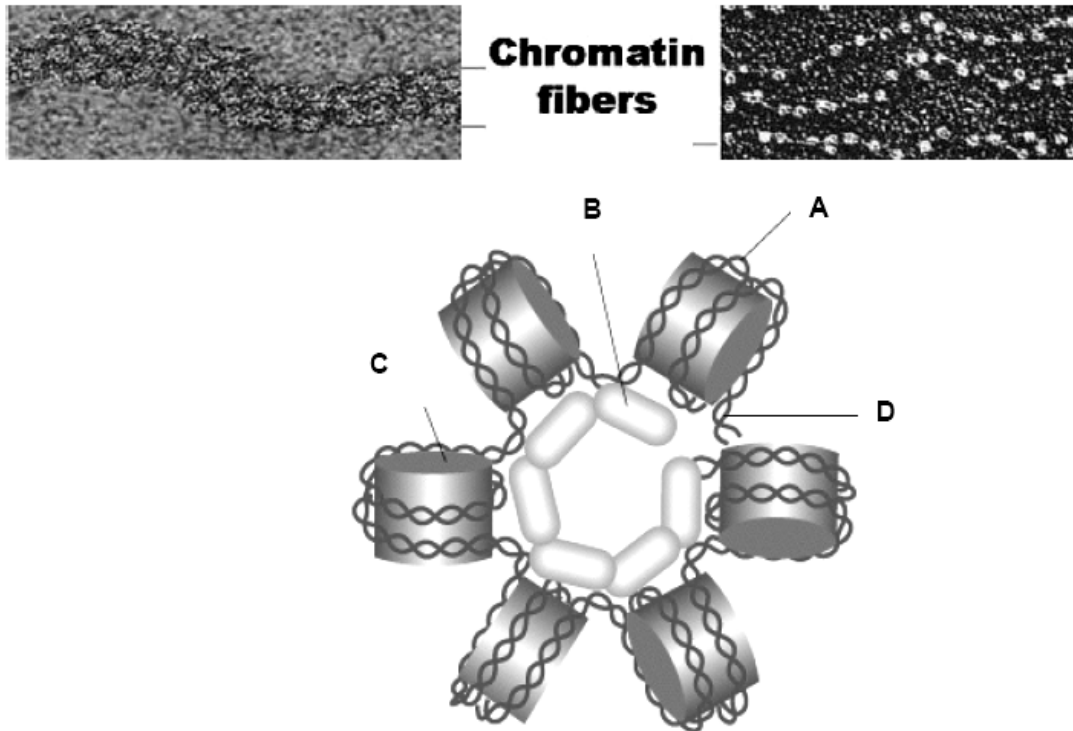
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[Qn 2 Total: /9]

QUESTION 3

The following diagram shows the DNA packing of chromatin fibres in a eukaryotic cell.



(a) Describe how DNA is organized to form chromosomes. [5]

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- (b)** Explain briefly how simultaneous modifications of DNA and histones can control gene expression. [2]

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Gene expression in eukaryotes requires the process of transcription and translation

- (c)** State how 2 such post transcriptional modifications may be carried out and explain the purpose of each modification. [4]

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- (d)** Suggest a type of post-translational modification and state why it may be necessary. [1]

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[Qn 3 Total: /12]

QUESTION 4

The flowers of the sweet pea plant are either purple or white. Flowers are purple if they contain anthocyanin pigment, and white if they do not. Two independently assorting genes, **C** and **P**, are involved in anthocyanin synthesis. Each gene has a recessive allele that abolishes pigment production (i.e. **cc** is epistatic over **P**, and **pp** is epistatic over **C**).

White flower Variety 1 has the genotype **CCpp** whilst white flower Variety 2 has the genotype **ccPP**. The two varieties are crossed to obtain F₁ offsprings, which all had purple flowers. The hybrids were then intercrossed to give a F₂ generation.

- (a) (i)** Draw a genetic diagram to show the gametes and the genotypes and phenotypes of the F₁ and F₂ generations of the two crosses. Give the ratio of phenotypes in the F₂ generation. [5]

(ii) Suggest the biochemical basis for this form of epistasis. [2]

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(b) The number of sweet pea plants with purple flowers and white flowers in a natural population was found to be 114 and 62 respectively. The chi-squared test was performed on these data.

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

where Σ = 'sum of...'
 ν = degrees of freedom
 c = number of classes

O = observed 'value'
 E = expected 'value'

A chi-squared table.

degrees of freedom	probability, p				
	10%	5%	2%	1%	0.1%
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(i) Calculate the value of χ^2 applicable to these data. (Workings need not be shown). [1]

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(ii) Use the χ^2 table provided to find the probability of that the difference between the expected and observed numbers and is due to chance. [1]

Probability

=

[Qn 4 Total: /9]

QUESTION 5

Type 2 diabetes is a disease condition that is usually caused by the failure of target cells to respond to insulin. This is due to a defect in the insulin response mechanism of the target cells.

The figure below shows the normal sequence of cell signaling events when insulin reaches its target cell and binds to an insulin receptor.

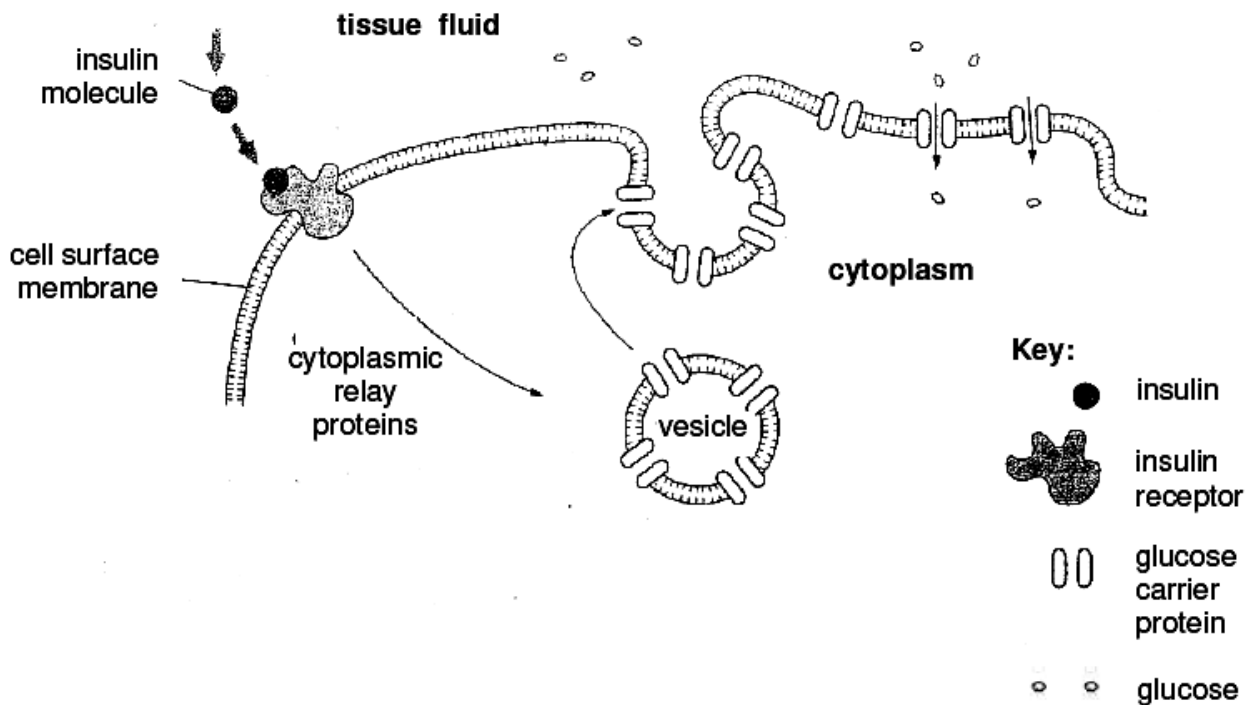


Fig 5.1

(a) State the source of the insulin molecule in the figure above.

[1]

The structure of the insulin receptor is shown in the figure below.



Fig. 5.2

(b) With reference to **Fig. 5.1** and **Fig. 5.2** above,

- (i)** explain how the binding of insulin to its receptor results in an increase in the phosphorylation of serine and threonine residue side-chains of the cytoplasmic relay proteins. [3]

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- (ii)** describe the events subsequent to the phosphorylation of serine and threonine residues of cytoplasmic relay proteins that lead to an increase in the number of glucose channels in the cell surface membrane. [3]

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The table below shows the relative levels of glucose and insulin in Mr Keanu Rives and Mr Brad Pity. Data was obtained for two conditions, during a 6 hour fast and immediately after a meal high in carbohydrates.

Mr Rives is medically normal with respect to his blood glucose levels. However, Mr Pity has abnormally high levels of blood glucose and has thus been diagnosed as a sufferer of diabetes.

	Blood Glucose Levels / Mg per 100ml	Insulin levels / arbitrary units
Mr Rives		
- Fasting	90	5
- After a meal	120	50
Mr Pity		
- Fasting	140	5
- After a meal	190	10

Table 5.3

(c) With reference to Table 5.3, explain why Mr. Pity is considered to be suffering from diabetes. [2]

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(d) Suggest reasons for the following:

(i) Excess blood glucose may cause frequent thirst. [2]

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(ii) Sufferers of diabetes tend to lose weight. [2]

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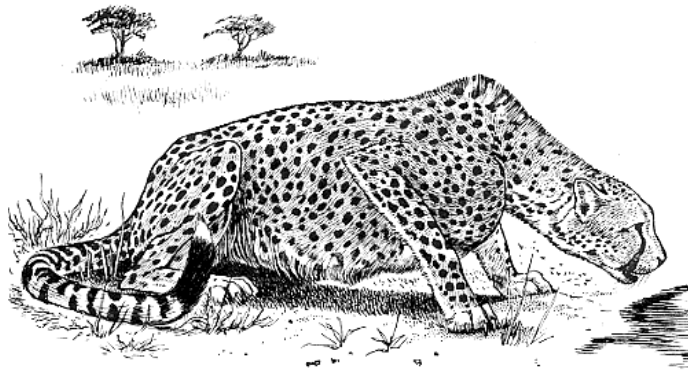
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[Qn 5 Total: /13]

QUESTION 6

The African cheetah, *Acinonyx jubatus* is an atypical member of the cat family (Felidae). It was suggested that the species had somehow lost its genetic variation. Physiological studies revealed that the populations of cheetahs are so closely related to each other such that skin grafts from one cheetah to another would not provoke immune response, thus suggesting an extreme population bottleneck in the past.



(a) State the genus to which the cheetah belongs.

[1]

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(b) Explain what is meant by a bottleneck event and how it can result in reduction in variation.

[3]

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(c) Suggest two problems that could result from the genetic uniformity faced by the South African cheetah population.

[2]

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- (d)** Suggest and explain how variation in the cheetah population can increase slowly but erratically. [3]

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- (e)** Describe how the molecular clock can be use to date evolutionary events such as how long since the cheetah species and the domestic cat species have diverged. [4]

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[Qn 6 Total: /13]

QUESTION 7

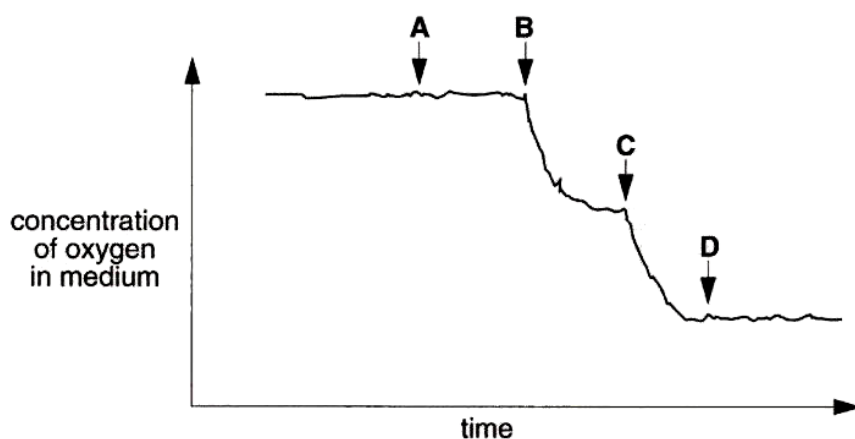
Aerobic respiration of glucose requires enzymes. The process is divided into four stages: glycolysis, the link reaction, Krebs cycle and oxidative phosphorylation.

- (a) Complete the table to indicate the **precise location** of the four stages. [2]

Respiratory Stage	Location in cell
Glycolysis	
Link reaction	
Krebs cycle	
Oxidative phosphorylation	

Liver cells are frequently used as a source of mitochondria. These cells are homogenised in a sucrose solution and the mitochondria isolated. The suspended organelles can be measured over a given time period.

The figure below shows a print-out from this apparatus. At this point **A** the respiratory substrate glucose was added. At points **B**, **C** and **D** equal quantities of ADP were added.



- (b) Explain the results,

- (i) between points **A** and **B** [4]

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(ii) between points **B** and **C**

[3]

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(iii) after point **D**

[3]

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[Qn 7 Total: /12]

Section B

*Answer **one** of the questions.*

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 8** **(a)** Contrast the mode of entry of influenza virus and HIV. [4]
 (b) Describe in detail the reproductive cycle of T4 phage. [9]
 (c) With reference to lambda phage and its reproductive cycle, outline how specialized transduction could result in a stable transfer of genes into a recipient bacterium. [7]

[Total: 20]

- 9** **(a)** Outline the roles of calcium, sodium and potassium ions in the transmission of impulses. [10]
 (b) Contrast the transmission of nerve impulse along a neurone and synaptic transmission. [6]
 (c) Explain how the temporal and spatial summation of various stimuli can result in a coordinated response at the postsynaptic neurone. [4]

[Total: 20]